VASANTHAGANESH R

Backend Java Engineer · Enterprise Microservices & Scalable Systems

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PROFESSIONAL SUMMARY

Backend Java Engineer with 2+ years of professional enterprise experience at **Virtusa** engineering high-throughput Spring Boot microservices and transactional banking workflows for **Citi's global banking platform**. Deep expertise in Core Java (8/17), ACID compliance, relational query optimization, and contract-first REST APIs where zero data loss and 99.99% availability are non-negotiable. **Oracle Certified Associate (OCA) Java Programmer** and **Claude AI Certified**, combining battle-tested BFSI engineering rigor with modern AI-augmented developer velocity and full-stack product execution.

TECHNICAL SKILLS

Backend & Core Java:	Core Java (8/17), Spring Boot, Spring Data JPA, Hibernate, Microservices, REST APIs, Concurrency, Maven, JUnit, Mockito
Databases & Integrity:	PostgreSQL, MySQL, Oracle DB, Schema Design, Indexing & Query Tuning, Strict ACID Compliance
AI & Emerging Tools:	Claude API, Anthropic Prompt Engineering, Context Memory Architectures, Python, LLM Workflows
Frontend & Mobile:	React, React Native, TypeScript, Node.js, Flutter, Cross-Platform Mobile Architecture
DevOps & 3D Tools:	Git, IntelliJ IDEA, Docker, Postman, Linux, Blender, Unity, 3D Spatial Systems

PROFESSIONAL EXPERIENCE

Associate Engineer — Backend Java | Virtusa (Client: Citi) — Aug 2023 – Sep 2025 | Puducherry, India

- Engineered high-concurrency **Spring Boot microservices** and secure REST endpoints powering Citi's global enterprise banking platform under strict BFSI compliance standards.
- Enforced strict **ACID transactional boundaries**, deterministic rollback handlers, and idempotent endpoints across financial transactions, guaranteeing **zero financial data loss** during network partitions.
- Optimized relational database queries, indexing strategies, and Hibernate/JPA object mappings on **Oracle DB**, mitigating query bottlenecks during peak clearing hours.
- Automated enterprise unit and integration test coverage using **JUnit and Mockito**, adhering to banking compliance, audit regulations, and continuous integration pipelines.
- Collaborated in distributed Agile sprints with global cross-functional squads to deliver mission-critical banking features.

Independent Backend & Systems Engineer | Self-Employed — Oct 2025 – Present | Remote

- **Precision Tools QC & Report Manager:** Designed and deployed an industrial mobile QC system in React Native with automated PDF generation, replacing manual shop-floor paper inspections with an automated **20-second digital verification workflow**.
- **Precision Tools B2B Corporate Platform:** Built a high-performance React B2B showcase featuring asset optimization, delivering sub-second load times and seamless commercial product catalogs.
- **Daily Vegetable Price Tracker:** Developed a community Android utility publishing real-time morning vegetable market rates across Puducherry, protecting household budgets from price gouging.
- Managed complete engineering lifecycles independently: client discovery, relational schema design, API architecture, implementation, and cloud deployment.

Technical Internships | VEI Technology & Askan Technology — 2022 | Puducherry, India

- **VEI Technology (IoT Systems):** Engineered a hardware-to-cloud sensor data ingestion pipeline and telemetry dashboard.
- **Askan Technology:** Completed in-plant training in cross-platform mobile application development with Flutter.

FEATURED SYSTEMS & AI PROJECTS

AI Reflection Journal React · TypeScript · AI Prompts · Mental Wellness · Privacy github.com/vasanthaganesh/AI-Reflection-Journal ↗

- Architected an introspective digital sanctuary for introverts and overthinkers who struggle to vocalize chaotic thoughts out loud.
- Engineered thoughtful prompt chains and private cognitive restructuring workflows, transforming silent overwhelm into structured clarity without judgment.

AI Companion Python · LLM Workflows · Persistent Context Memory · TypeScript github.com/vasanthaganesh/AI_Companion ↗

- Engineered an empathetic conversational partner equipped with **persistent contextual memory** across sessions to combat social isolation.
- Retains personal nuances and conversational history over time, delivering an attentive, safe sounding board and meaningful presence.

Citi Enterprise Banking Platform Java 8/17 · Spring Boot · Microservices · Oracle DB · REST APIs Virtusa Client Project

- Core transactional banking pipelines handling high-throughput institutional and retail transactions under strict financial audit standards.

ACADEMIC & ENGINEERING PROJECTS

3D Interactive Virtual Campus Visualization Blender · Unity · 3D Modeling · Interactive Systems · C# Final Year Capstone Project

- Architected and developed a full-scale **3D interactive virtual campus exploration platform** showcasing the complete architectural infrastructure of the college.
- Engineered complex 3D building and campus models with optimized polygon budgets, realistic material textures, custom collision-detection camera navigation, and spatial point-of-interest triggers for an immersive real-time user walkthrough.

Appverse Application Android · Java · Firebase Realtime DB Campus Academic Portal

- Campus collaboration mobile application facilitating assignment submissions, real-time departmental announcements, and professor-student notifications.

AudioBook Application Android · Java · Accessibility APIs Assistive Mobile App

- Accessibility-focused audiobook application tailored for learning-disabled students and second-language learners with offline audio caching and variable speed playback.

Abstraction-Based Text Summarization Python · NLP · Flask · Web Application NLP Analytical Tool

- NLP-based abstractive text summarizer developed to assist investigation officers in digesting voluminous investigative reports into concise, structured intelligence briefs.

CERTIFICATIONS

- **Oracle Certified Associate (OCA)**, Java SE 8 Programmer (Core Java & Concurrency) — Flagship Credential
- **Claude AI Certified** — Anthropic Prompt Engineering & Generative Workflows
- **Talent-Next Java Full Stack Certification** — Wipro / NASSCOM
- **JavaScript Essentials 1** — Online Certification
- **Linux Training** — Spoken Tutorial, IIT Bombay

HONORS & HACKATHONS

- 🥉 **3rd Prize** — AI-HACK Project Development Competition (MVIT)
- 🥉 **3rd Prize** — SCIMIT-23 National Level Project Expo
- 🥇 **1st Prize** — Paper Presentation, Christ College of Engineering
- 🥇 **1st Prize** — Blind Coding, Pondicherry Technological University
- 🏅 **Participant** — Smart India Hackathon (SIH 2022) & Virtusa Jatayu

EDUCATION & VITALS

Bachelor of Technology (B.Tech) — Information Technology | Manakula Vinayagar Institute of Technology Puducherry, India | Graduated 2023 (CGPA: 8.32 / 10.0)

Languages: English, Tamil • **Availability:** Immediate • **Relocation:** Open to relocation globally